Decision and Risk Analysis

Lecture 1: **Bayesian Thinking for Risk and Decision Analysis**

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A decision problem begins with a **choice to be made** under certain **objectives** and **constraints**.

Risk and Uncertainty

- **Choice under risk**: Outcome probabilities are known (or can be estimated reliably).
- **Choice under uncertainty**: Outcome probabilities are unknown (or difficult to estimate).

Decision and Risk Analysis

1. **Modeling Decisions**: Define the available choices or actions.
2. **Modeling Risk**: Analyze situations where outcome probabilities are known.
3. **Modeling Uncertainty**: Analyze situations where outcome probabilities are unknown (or difficult to estimate).
4. **Modeling Preferences**: Represent how decision makers value outcomes and trade-offs.

Bayesian Modeling for Decision Analysis

- Bayesian modeling provides a way to update beliefs as new information becomes available.
- It combines:
    1. **Prior beliefs**: what we know or assume before seeing new data.
    2. **Evidence/data**: new information relevant to the decision.
    3. **Posterior beliefs**: updated probabilities after incorporating the evidence.
- In decision analysis, Bayesian modeling helps decision makers revise risk estimates and make better choices under uncertainty.
- In short, it supports learning from evidence and improving decisions over time.